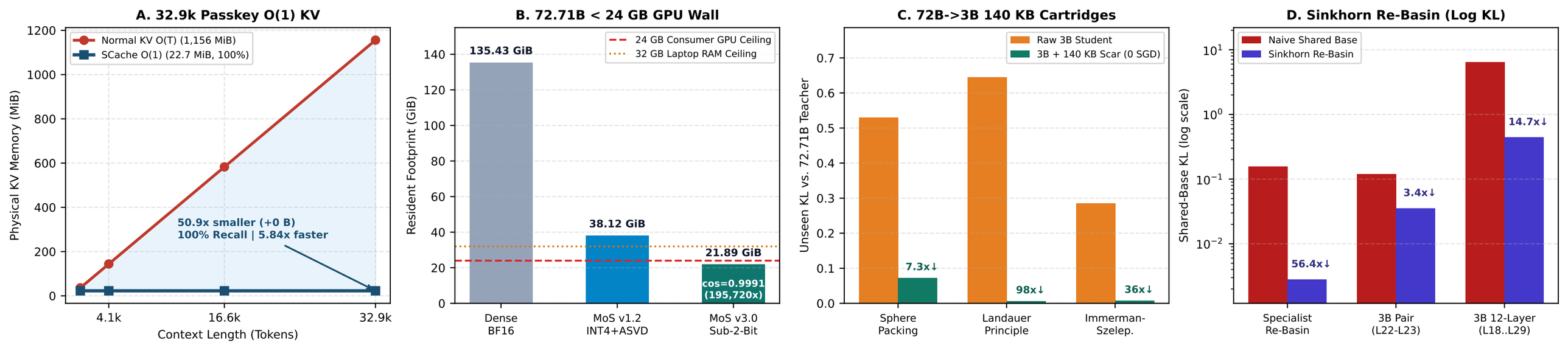
A. 32.9k Passkey O(1) KV
Normal KV O(T) (1,156 MiB)
SCache O(1) (22.7 MiB, 100%)
50.9x smaller (+0 B)
100% Recall | 5.84x faster
Physical KV Memory (MiB)
Context Length (Tokens)
4.1k
16.6k
32.9k
B. 72.71B < 24 GB GPU Wall
24 GB Consumer GPU Ceiling
32 GB Laptop RAM Ceiling
135.43 GiB
38.12 GiB
21.89 GiB
cos=0.9991
(195,720x)
Resident Footprint (GiB)
Dense
BF16
MoS v1.2
INT4+ASVD
MoS v3.0
Sub-2-Bit
C. 72B->3B 140 KB Cartridges
Raw 3B Student
3B + 140 KB Scar (0 SGD)
7.3x↓
98x↓
36x↓
Unseen KL vs. 72.71B Teacher
Sphere
Packing
Landauer
Principle
Immerman-
Szelep.
D. Sinkhorn Re-Basin (Log KL)
Naive Shared Base
Sinkhorn Re-Basin
56.4x↓
3.4x↓
14.7x↓
Shared-Base KL (log scale)
Specialist
Re-Basin
3B Pair
(L22-L23)
3B 12-Layer
(L18..L29)